Academic Requirements

Syllabus, Deadlines, Report Structure, Professor Feedback

March 31, 2026

ClawCraft Project Reference

Project Type

Option A: Generative AI-driven art project using at least two different media types and architectures, accompanied by a written report.

Deliverables & Grades

1. Final Project Proposal -- Due March 19 (Moodle). 10% of final grade. SUBMITTED AND GRADED. Was 5 pages (professor noted rubric said 3).
2. Final Project Presentation -- April 9 in class, recorded. 20% of final grade.
3. Final Project Report -- Due April 16, submit online. 20% of final grade.

Presentation Requirements (April 9)

- Total duration: 12 minutes STRICT MAXIMUM
- Project description: 6 minutes to explain objective and implementation
- Demonstration: 6 minutes to showcase project in action
- Audience: Assume NOT familiar with Generative AI or ML. Provide high-level intro to concepts.
- Emphasize what makes the project interesting, including both successes and failures
- Rehearse and test all equipment beforehand
- Submit presentation as PDF on Moodle

Report Requirements (April 16)

Format

- PDF document
- 2,000 to 3,000 words (excluding abstract, references, appendices)
- Chicago author-date citation style
- Recommended: ACM HCI LaTeX Template (mandatory only for research paper option)
- Include diagrams, images, visuals. Do NOT include code screenshots -- link to GitHub instead.
- Include link to short demo video documenting the project
- Include links to code repository and deployed web app (if applicable)
- Audience: Assume reader IS knowledgeable in ML and AI. Explain technical aspects in detail.

Required Report Structure

1. Abstract (150-200 words): Brief summary -- purpose, methodology, outcomes.
2. Introduction
    - Context: Situate project within broader field

- Literature review: Reference similar projects and relevant academic work
- Significance: Explain importance or innovative aspects
- Objectives: Clearly state what project aims to achieve

3. Project Description
   - Vision for generative AI in creative practice: conceptual approach, how GenAI tools support the project idea
   - Technologies used: software tools and frameworks, justified
   - ML tasks: specific ML tasks implemented (text-to-image, text-to-speech, conditional generation, denoising, etc.)
   - Infrastructure: computational resources and platforms used
   - Data: how gathered/sourced, size, number of classes, data points
   - System Integration: how all components work together cohesively
4. Process
   - Development stages: steps from proposal to final delivery
   - Challenges and solutions: obstacles encountered and how addressed
   - Successes: what worked well and unexpected positive outcomes
   - Learning experiences: key insights and skills gained
5. Future Work: potential improvements or expansions for future iterations
6. References: Chicago author-date style, all sources cited
7. Appendix: demo video link, code repo link, deployed app link. NOT included in word count.

Rubric Requirements for ClawCraft

Two Media Types Required

1. 3D models / rendered scenes / short animated sequences (Blender EEVEE renders)
2. Images (DALL-E concept art + Blender renders)

Two ML Architectures Required

1. Transformer: GPT-5.3-Codex for natural language -> Blender Python command generation
2. Diffusion/Autoregressive: gpt-image-1 (DALL-E) for concept art generation

CRITICAL: DALL-E has NOT been used yet. This MUST happen before the presentation.

ML Tasks to Document

- Text-to-3D via code generation (Codex writes bpy Python from natural language)
- Text-to-image (DALL-E concept art)
- Iterative conditional generation (prompt refinement feedback loop)

Data to Document

- render_log.json -- auto-collected per render (prompt, commands, timing, outcome, observation)

- workflow_notes.md -- manual process observations
- Notion databases (Scenes, Renders, Tasks, Milestones)
- Discord chat history -- prompt/response threads per scene
- Render file corpus (/tmp/clawcraft/*.png)

Professor Feedback (6 exact points from Vig)

These are the professor's exact comments on the proposal. ALL must be addressed in the final report.

1. "Misleading at first" -- The overview makes it sound like magic. Say upfront it's an iterative feedback loop: prompt -> evaluate -> refine -> repeat. First paragraph. Don't bury it in the scene building section.
2. "Objective needs revision" -- The goal isn't just "make art pieces." It's to compare human vs AI creative workflows and document what you learn. The art is the output; the insight is the contribution. He wants: what are the different affordances of AI-assisted 3D creation vs doing it by hand?
3. "Artistic direction with prompts" -- He wants actual planned prompts tied to artistic vision. Not generic descriptions. Specific: "scenes from childhood game memories: a blocky hallway, a kitchen with impossible geometry, a fridge interior that extends into a landscape."
4. "Measurable milestones" -- Concrete weekly deliverables. Not "begin artistic exploration." Instead: "Week 2: generate 5 test objects. Week 3: produce 3 complete scenes with iterative refinement."
5. "Literature review" -- Engage critically. Don't just list sources. Say what they found and how it relates. Example: "Ahuja (2025) demonstrated MCP enables two-way Blender control, but noted geometry cleanup was often needed."
6. "Concise" -- Proposal was 5 pages, rubric said 3. Report: stay within 2000-3000 words.

What the Professor Actually Wants

A section in the report reflecting on what the ClawCraft workflow ENABLES vs LIMITS compared to doing it manually. The comparison between AI-assisted and hand-built 3D creation is the academic contribution. That's writing, not infrastructure.

Operational Implications

- Document the feedback loop, not just the output. The Discord conversation IS the data.
- Compare to hand-building when possible. Note what was faster, better, worse, surprising.
- The comparison is what the professor wants as the actual contribution -- not just 'look what AI made.'

Report Checklist

- [ ] Iterative loop stated upfront in introduction (feedback point 1)
- [ ] Objective includes workflow insights -- human vs AI affordances (feedback point 2)
- [ ] Successes AND failures documented explicitly (feedback point 3)
- [ ] Specific artistic direction with planned prompts (feedback point 4)

- [ ] Measurable milestones with weekly deliverables (feedback point 5)
- [ ] Literature engaged critically, not just listed (feedback point 6)
- [ ] Within 2000-3000 words (feedback point 7)
- [ ] Demo video link included
- [ ] GitHub repo link included
- [ ] DALL-E concept art generated and documented (second ML architecture)
- [ ] Chicago author-date citations throughout
- [ ] No code screenshots -- GitHub links instead

References (from proposal)

- Ahuja 2025 -- BlenderMCP: demonstrated MCP enables two-way Blender control, noted geometry cleanup often needed
- Lima 2025 -- 3D-Agent
- Eliason 2026 -- FelixAI / OpenClaw agent
- Ho et al 2020 -- DDPM / denoising diffusion probabilistic models
- Vaswani et al 2017 -- Attention Is All You Need (transformer architecture)
- Ouyang et al 2022 -- RLHF / InstructGPT
- Yao et al 2022 -- ReAct: reasoning + acting in LLMs
- Bitdefender 2026, Cisco 2026, Microsoft 2026 -- OpenClaw security advisories